
O-RAN Work Group 10 (OAM for O-RAN)

O-RAN Information Model and Data Models Specification

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Foreword

This Technical Specification (TS) has been produced by WG10 of the O-RAN ALLIANCE.

The content of the present document is subject to continuing work within O-RAN and may change following formal O-RAN approval. Should the O-RAN Alliance modify the contents of the present document, it will be re-released by O-RAN with an identifying change of version date and an increase in version number as follows:

version xx.yy.zz

where:

- xx: the first digit-group is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc. (the initial approved document will have xx=01). Always 2 digits with leading zero if needed.
- yy: the second digit-group is incremented when editorial only changes have been incorporated in the document. Always 2 digits with leading zero if needed.
- zz: the third digit-group included only in working versions of the document indicating incremental changes during the editing process. External versions never include the third digit-group. Always 2 digits with leading zero if needed.

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the O-RAN Drafting Rules (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are **NOT** allowed in O-RAN deliverables except when used in direct citation.

1 Scope

This document is both a Specification and an Informational Report in that it specifies the Information Model and the Data Models that are foundational for O-RAN's model-driven architecture and for the functions carried out over O-RAN interfaces.

In addition, this document includes information about existing standards and industry work that serve as a basis for work items in O-RAN. There is a "prosumer" relationship evolving between O-RAN and 3GPP, as each makes its model available and provides model feedback to the other.

Lastly, this document describes the de facto methods and procedures with respect to a "modeling continuum" that aims to establish and evolve an O-RAN Information Model from which O-RAN Data Models may be generated manually or with a set of tools.

2 References

2.1 Normative references

References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific. For a specific reference, subsequent revisions do not apply. For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document, a non-specific reference implicitly refers to the latest version of that document in Release 18, or the latest 3GPP release prior to Release 18 that includes that document.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, O-RAN cannot guarantee their long-term validity.

The following referenced documents are necessary for the application of the present document.

- [1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications"
- [2] 3GPP TS 28.622: "Telecommunication management; Generic Network Resource Model (NRM) Integration Reference Point (IRP); Information Service (IS)"
- [3] RFC 8342: "Network Management Datastore Architecture (NMDA)", IETF
- [4] RFC 8407: "Guidelines for Authors and Reviewers of Documents Containing YANG Data Models", IETF, October 2018
- [5] RFC 7950, "The YANG 1.1 Data Modeling Language", IETF
- [6] 3GPP TS 32.156: "Telecommunication management; Fixed Mobile Convergence (FMC) Model repertoire"
- [7] 3GPP TS 32.160: "Management and orchestration; Management service template"

- [8] O-RAN.WG10.TS.O1-Interface: “O-RAN O1 Interface Specification”
- [9] OMG formal/2017-12-05: OMG® Unified Modeling Language® (OMG UML®), Version 2.5.1
- [10] O-RAN.WG3.TS.O1-Interface-for-Near-RT-RIC: “O1 Interface Specification for Near Real Time RAN Intelligent Controller”
- [11] 3GPP TS 28.625: “Telecommunication management; State Management Data Definition Integration Reference Point (IRP); Information Service (IS)”
- [12] O-RAN.WG6.TS.O2IMS-INTERFACE: “O2ims Interface Specification”
- [13] 3GPP TS 32.404: “Technical Specification Group Services and System Aspects; Telecommunication management; Performance Management (PM); Performance measurements”
- [14] ETSI GS NFV-IFA 027 V4.2.1, ”Network Functions Virtualisation (NFV) Release 4; Management and Orchestration; Performance Measurements Specification”
- [15] 3GPP TS 28.620: “Telecommunication management; Fixed Mobile Convergence (FMC), Federated Network Information Model (FNIM), Umbrella Information Model (UIM)”
- [16] 3GPP TS 28.111: “Management and orchestration; Fault Management (FM)”

2.2 Informative references

References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific. For a specific reference, subsequent revisions do not apply. For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document, a non-specific reference implicitly refers to the latest version of that document in Release 18, or the latest 3GPP release prior to Release 18 that includes that document.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, O-RAN cannot guarantee their long-term validity.

The following referenced documents are not necessary for the application of the present document, but they assist the user with regard to a particular subject area.

- [i.1] O-RAN.WG4.TS.MP.0: “O-RAN Alliance Working Group 4 Management Plane Specification”
- [i.2] RFC 6241, “Network Configuration Protocol (NETCONF)”, IETF, June 2011
- [i.3] “Modeling, Use Case and Architecture Process,” B. Cheung et al, May 2019, ONAP
- [i.4] O-RAN.WG2.TS.A1AP: “O-RAN Working Group 2 (Non-RT RIC and A1 interface WG) A1 interface: Application Protocol”
- [i.5] O-RAN.WG3.TS.E2GAP: “O-RAN Working Group 3 Near-Real-time RAN Intelligent Controller Architecture & E2 General Aspects and Principles”
- [i.6] O-RAN-WG5.O-CU-O1: “O1 Interface specification for O-CU-UP and O-CU-CP”
- [i.7] O-RAN.WG5.O-DU-O1: “O1 Interface specification for O-DU”
- [i.8] 3GPP TS 28.623: “Telecommunication management; Generic Network Resource Model (NRM) Integration Reference Point (IRP); Solution Set (SS) definitions”
- [i.9] 3GPP TS 36.423: “Technical Specification Group Radio Access Network; Evolved Universal Terrestrial Radio Access Network (E-UTRAN); X2 application protocol (X2AP)”

- [i.10] 3GPP TS 38.460: “Technical Specification Group Radio Access Network; NG-RAN; E1 general aspects and principles”
- [i.11] 3GPP TS 38.470: “Technical Specification Group Radio Access Network; NG-RAN; F1 general aspects and principles”
- [i.12] 802.1X-2020, “IEEE Standard for Local and Metropolitan Area Networks--Port-Based Network Access Control”, IEEE, 2020-02-28
- [i.13] RFC 5717: “Partial Lock Remote Procedure Call (RPC) for NETCONF”, IETF, December 2009
- [i.14] RFC 6022: “YANG Module for NETCONF Monitoring”, IETF, October 2010
- [i.15] RFC 6243: “With-defaults Capability for NETCONF”, IETF, June 2011
- [i.16] RFC 6470: “Network Configuration Protocol (NETCONF) Base Notifications”, IETF, February 2012
- [i.17] RFC 6643: “Translation of Structure of Management Information Version 2 (SMIv2) MIB Modules to YANG Modules”, IETF, July 2012
- [i.18] RFC 6991: “Common YANG Data Types”, IETF, July 2013
- [i.19] RFC 7224: “IANA Interface Type YANG Module”, IETF, May 2014
- [i.20] RFC 7317: “A YANG Data Model for System Management”, IETF, August 2014
- [i.21] RFC 7758: “Time Capability in NETCONF”, IETF, February 2016
- [i.22] RFC 7952: “Defining and Using Metadata with YANG”, IETF, August 2016
- [i.23] RFC 8072: “YANG Patch Media Type”, IETF, February 2017
- [i.24] RFC 8341: “Network Configuration Access Control Model”, IETF, March 2018
- [i.26] RFC 8343: “A YANG Data Model for Interface Management”, IETF, March 2018
- [i.27] RFC 8344: “A YANG Data Model for IP Management”, IETF, March 2018
- [i.28] RFC 8348: “A YANG Data Model for Hardware Management”, IETF, March 2018
- [i.30] RFC 8525: “YANG Library”, IETF, March 2019
- [i.31] RFC 8526: “NETCONF Extensions to Support the Network Management Datastore Architecture”, IETF, March 2019
- [i.32] RFC 8528: “YANG Schema Mount”, IETF, March 2019
- [i.33] RFC 8575: “YANG Data Model for the Precision Time Protocol (PTP)”, IETF, May 2019
- [i.34] RFC 8632: “A YANG Data Model for Alarm Management”, IETF, September 2019
- [i.35] RFC 8641: “Subscription to YANG Notifications for Datastore Updates”, IETF, September 2019
- [i.36] draft-ietf-netconf-crypto-types: “YANG Data Types and Groupings for Cryptography”, IETF
- [i.37] 3GPP TS 28.541: “Management and orchestration; 5G Network Resource Model (NRM); Stage 2 and stage 3”
- [i.38] RFC 3198: “Terminology for Policy-Based Management” , <https://datatracker.ietf.org/doc/html/rfc3198>
- [i.39] SemVer: “Semantic Versioning 2.0.0.”, <https://semver.org>
- [i.40] O-RAN.WG6.TS.O-CLOUD-IM: “O-Cloud Information Model”

3 Definitions of term, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms and definitions given in 3GPP TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in 3GPP TR 21.905 [1].

Information Model: See Information Model in RFC 3198 [i.38].

Data Model: See Data Model in RFC 3198 [i.38].

Model: See Model [Class] in UML [9], clause 12.4.4.

Data Dictionary: a centralized repository of information about data such as meaning, relationships to other data, origin, usage, and format.

Namespace: See namespace as specified in UML v2.5.1 [9], clause 7.8.10 Namespace [Abstract Class].

Primary Key: value which is distinctive for each record of the table.

Key value pairs: a set of data consists of two related data elements, key which is constant describing data set and its value.

3.2 Symbols

For the purposes of the present document, the symbols given in 3GPP TR 21.905 [1] and the following apply. A symbol defined in the present document takes precedence over the definition of the same symbol, if any, in 3GPP TR 21.905 [1].

3.3 Abbreviations and acronyms

For the purposes of the present document, the abbreviations given in 3GPP TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in 3GPP TR 21.905 [1].

3GPP	3 rd Generation Partnership Project
API	Application Programming Interface
CR	Change Requests
EMS	Element Management System
FCAPS	Fault, Configuration, Accounting, Performance, Security
IOC	Information Object Class
LS	Liaison Statement
MA	Managed Application
MANO	Management and Orchestration
ME	Managed Element
MF	Managed Function
MnS	Management Service
MO	Managed Object
MOC	Managed Object Class
MOI	Managed Object Instance

NAT	Network Address Translation
Near-RT RIC	O-RAN near real time RAN Intelligent Controller
NMS	Network Management System
Non-RT RIC	O-RAN non real time RAN Intelligent Controller
NRM	Network Resource Model
O-CU-CP	O-RAN Central Unit – Control Plane.
O-CU-UP	O-RAN Central Unit – User Plane
O-DU	O-RAN Distributed Unit
OMG	Object Management Group
O-RAN	Open Radio Access Network
O-RU	O-RAN Radio Unit
ONAP	Open Network Automation Platform
PNF	Physical Network Function
RAN	Radio Access Network
RRH	Remote Radio Head
SDO	Standards Development Organization
SMO	Service Management and Orchestration (layer)
TR	Technical Report
TS	Technical Specification
UML	Unified Modeling Language
SA5	Services & System Aspects Working Group 5
Telecom Management	
VNF	Virtualized Network Function

4 Overview and Philosophy

4.1 Relationship between O-RAN and Standards Development Organizations

The O-RAN Alliance complements the work of other Standards Development Organizations (SDO): 3GPP, IETF, and IANA are among the primary sources for Management specifications for O-RAN components.

3GPP has published:

- its Generic Network Resource Model (NRM), including:
 - a Stage 2 NRM Information Model, in 3GPP TS 28.622 [2];
 - corresponding Stage 3 Solution Sets, one of which consists of YANG Data Models, in 3GPP TS 28.623 [i.8];
- its 5G Network Resource Model (NRM), including the Stage 2 NR NRM Information Model and corresponding Stage 3 Solution Sets, one of which consists of YANG Data Models, in 3GPP TS 28.541 [i.37];
- “Management and Orchestration APIs”, including YANG data models from both NRMs, in a publicly available git repository

O-RAN Information Model, Data Models, and modeling processes should complement the work of other SDOs, not conflict or compete.

When 3GPP makes updates to their specification, they are incorporated into the O-RAN OAM Interface Specification [8]. This, in turn, drives the updates that are to be made to both the Information Model and Data Models within O-RAN.

5 O-RAN Information Model

5.1 Information Modeling Conventions

5.1.1 Modeling approach, Unified Modeling Language (UML)

The Information Model within O-RAN shall use the Unified Modeling Language™ (UML®) version 2.5.1 [9] from the Object Management Group (OMG).

O-RAN Information Model shall follow methodology documented in 3GPP TS 32.160 [7] Clause 5.2.

5.1.2 Namespaces in the Information Model

A Namespace provides a container for named elements, which are called its members. A Namespace may also import named elements from other specifications, in which case those elements become members of the importing Namespace.

5.2 O-RAN Information Model Definitions

5.2.1 Namespace: ORAN.ManagementArtifacts

5.2.1.1 Namespace Overview

The O-RAN Management Artifacts namespace identifies classes required to represent documentation descriptions that supplement the management aspect of elements represented by the classes in the NRM.

5.2.1.2 Imported and associated information entities

5.2.1.2.1 Imported information entities and local labels

Label reference	Local label
3GPP TS 28.622 [2], IOC, Top	Top

5.2.1.2.2 Associated information entities and local labels

Label reference	Local label
WG6.O2-Infrastructure Management Specification [12], IOC, ResourceType	OcloudResourceType

5.2.1.3 Class diagrams

5.2.1.3.1 Relationships

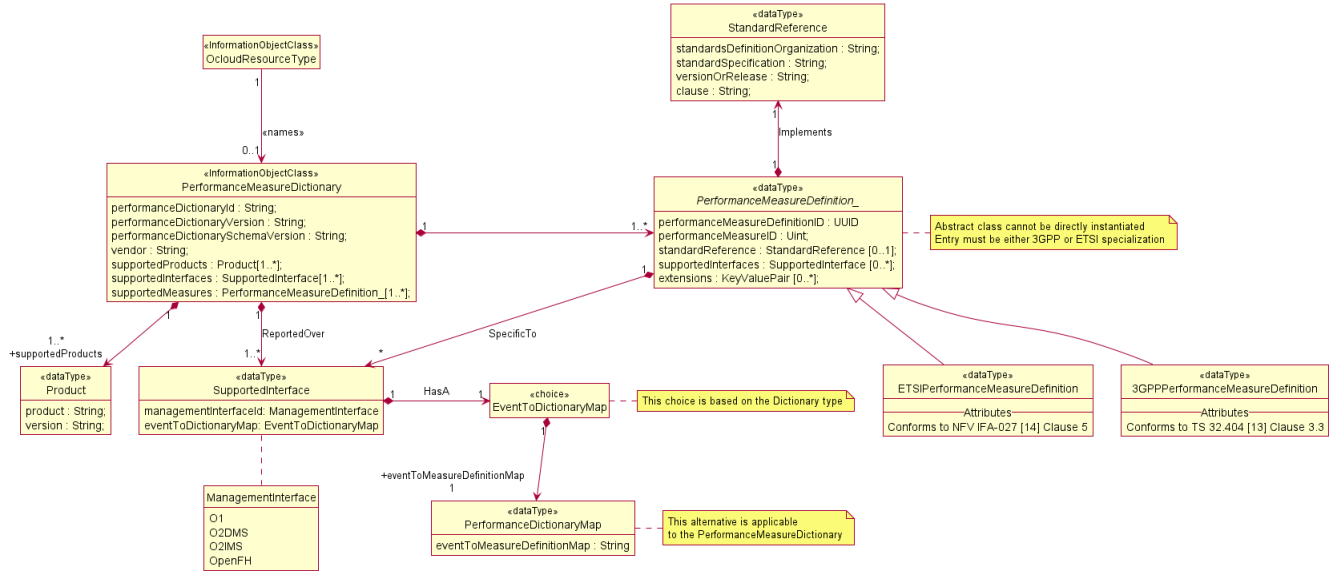


Figure 5.2.1.3.1-1 Performance Dictionary Relationships

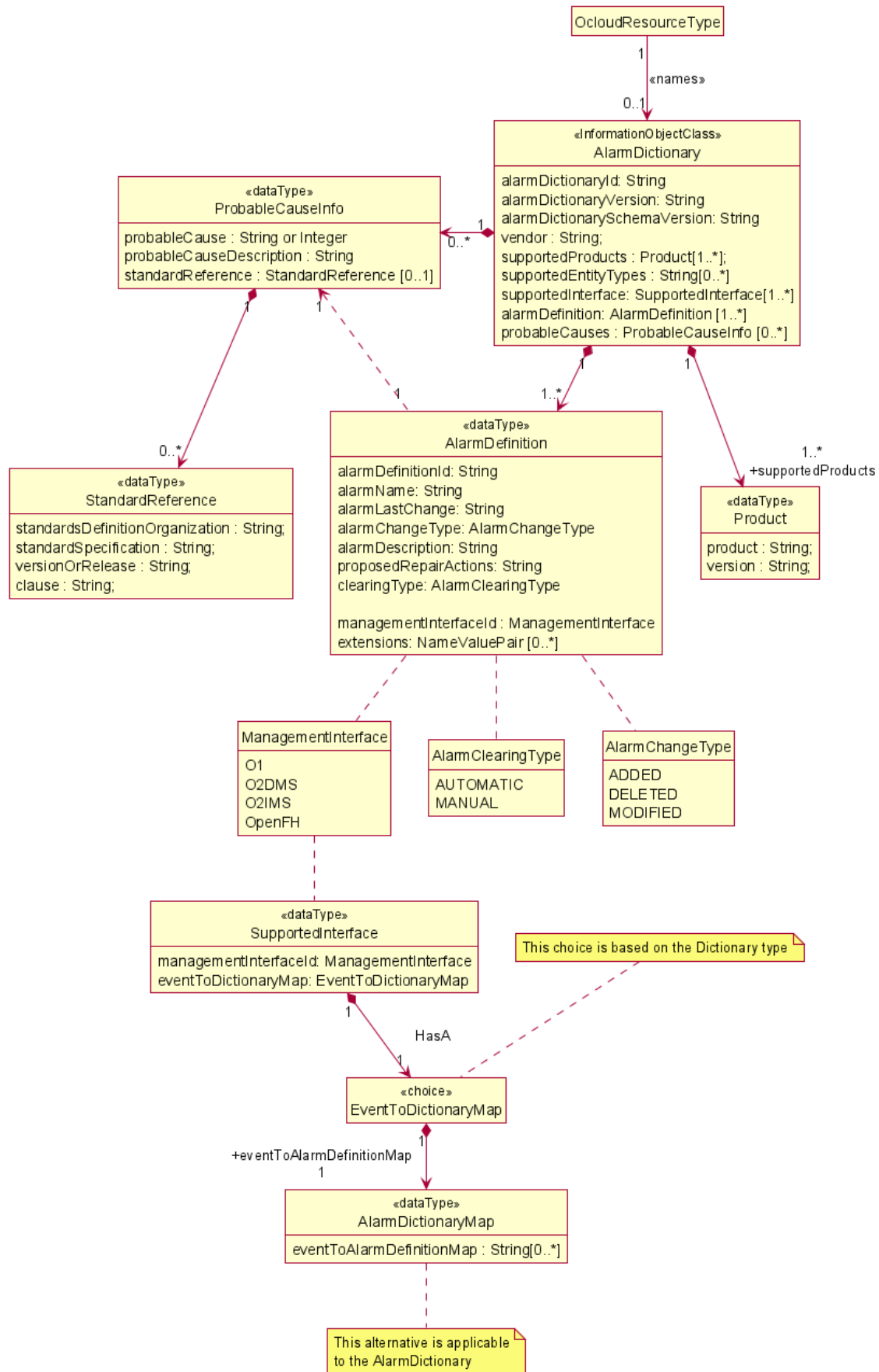


Figure 5.2.1.3.1-2 Alarm Dictionary Relationships

5.2.1.3.2 Inheritance

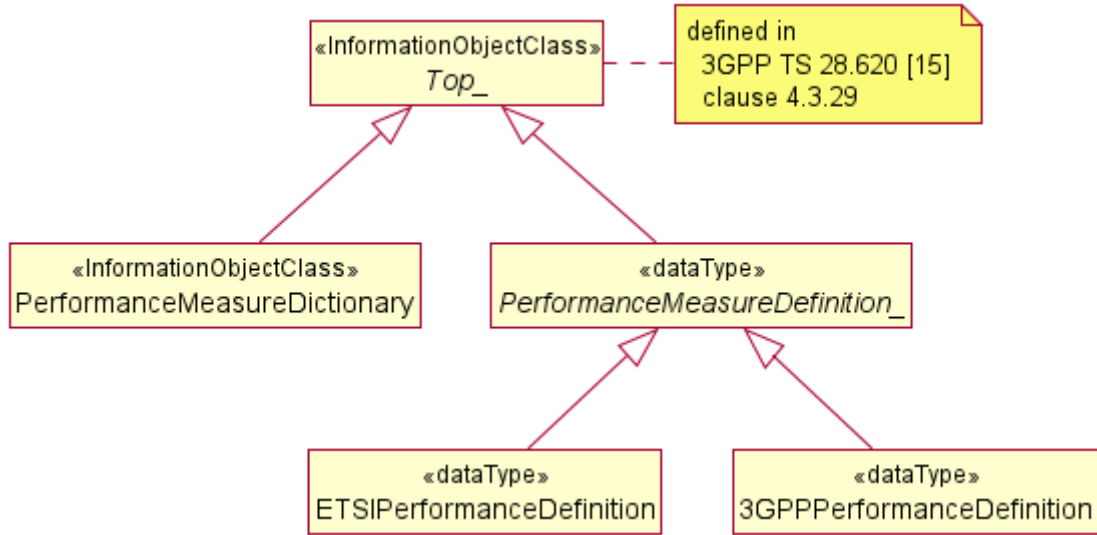


Figure 5.2.1.3.2-1 PerformancemeasureDictionary inheritance

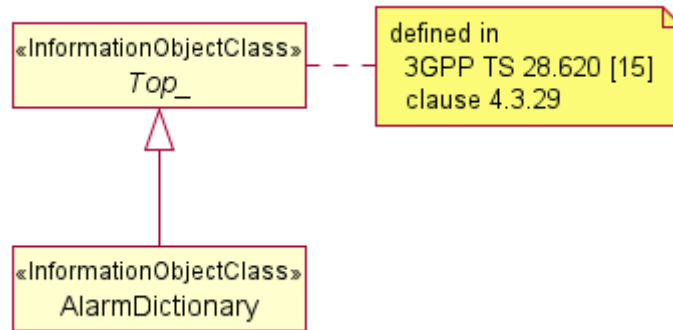


Figure 5.2.1.3.2-2 Alarm Dictionary inheritance

5.2.1.4 Class definitions

5.2.1.4.1 PerformanceMeasureDictionary

5.2.1.4.1.1 Definition

Resource Types in the O-Cloud provide the SMO a dictionary of performance measures available by Resources of the Resource Type. The O-Cloud Deployment Management Services also report performance of workloads deployed into the cloud. The DMS Performance follows ETSI-NFV specification IFA027 [14]. The performance dictionary should be structured

to allow resource vendors to specify their unique performance measures while also supporting performance measures which are defined by a standard.

5.2.1.4.1.2 Attributes

The PerformanceMeasureDictionary includes attributes inherited from Top IOC (defined in 3GPP TS 28.622 [2], clause 4.3.29) and the following attributes:

Attribute name	S	isReadable	isWritable	isInvariant	isNotifiable
performanceDictionaryId	M	T	F	T	F
performanceDictionaryVersion	M	T	F	T	F
performanceDictionarySchemaVersion	M	T	F	T	F
vendor	M	T	F	T	F
supportedProducts	M	T	F	T	F
supportedInterfaces	M	T	F	T	F
supportedMeasures	M	T	F	T	F

5.2.1.4.1.3 Attribute constraints

None

5.2.1.4.1.4 Notifications

This class does not support any notification.

5.2.1.4.1.5 State diagram

None

5.2.1.4.2 PerformanceMeasureDefinition_ <<dataType>>

5.2.1.4.2.1 Definition

The PerformanceMeasureDefinition defines the O-RAN attributes common to both variants of performance measurement definition.

5.2.1.4.2.2 Attributes

The PerformanceMeasureDefinition inherits attributes from Top IOC (defined in 3GPP TS 28.622 [2], clause 4.3.29) and extends it with the following attributes:

Attribute name	S	isReadable	isWritable	isInvariant	isNotifiable
performanceMeasureDefinitionId	M	T	F	T	F
performanceMeasureId	M	T	F	T	F
standardReference	M	T	F	T	F
supportedInterfaces	M	T	F	T	F
extensions	M	T	F	T	F

5.2.1.4.2.3 Attribute constraints

None

5.2.1.4.2.4 Notifications

This class does not support any notification.

5.2.1.4.2.5 State diagram

None

5.2.1.4.3 ETSIPerformanceMeasureDefinition <<dataType>>

5.2.1.4.3.1 Definition

The ETSIPerformanceMeasureDefinition is an extension of the PerformanceMeasureDefinition_ abstract class. It provides extended attributes specific to an ETSI based performance measurement definition.

5.2.1.4.3.2 Attributes

The ETSIPerformanceMeasureDefinition is a specialized class of the PerformanceMeasureDefinition_ class which inherits attributes from. It extends the class with the attributed defined in ETSI NFV IFA027 [14] clause 5.

5.2.1.4.3.3 Attribute constraints

None

5.2.1.4.3.4 Notifications

This class does not support any notification.

5.2.1.4.3.5 State diagram

None

5.2.1.4.4 3GPPPerformanceMeasureDefinition <<dataType>>

5.2.1.4.4.1 Definition

The 3GPPPerformanceMeasureDefinition is an extension of the PerformanceMeasureDefinition_ abstract class. It provides extended attributes specific to a 3GPP based performance measurement definition.

5.2.1.4.4.2 Attributes

The 3GPPPerformanceMeasureDefinition is a specialized class of the PerformanceMeasureDefinition_ class which inherits attributes from. It extends the class with fields defined in 3GPP TS 32.404 [13] clause 3.3.

5.2.1.4.4.3 Attribute constraints

None

5.2.1.4.4.4 Notifications

This class does not support any notification.

5.2.1.4.4.5 State diagram

None.

5.2.1.4.5 StandardReference <<dataType>>

5.2.1.4.5.1 Definition

When definition has been provided by another Standards body this dataType provides a reference to the standard body that is author of particular definition, as well as to where the definition can be found.

5.2.1.4.5.2 Attributes

The StandardReference contains attributes used to identify where in a standard the definition is provided.

Attribute name	S	isReadable	isWritable	isInvariant	isNotifiable
standardDefinitionOrganization	M	T	F	T	F
standardSpecification	M	T	F	T	F
versionOrRelease	M	T	F	T	F
clause	M	T	F	T	F

5.2.1.4.5.3 Attribute constraints

None

5.2.1.4.5.4 Notifications

This class does not support any notification.

5.2.1.4.5.5 State diagram

None.

5.2.1.4.6 SupportedInterface <<dataType>>

5.2.1.4.6.1 Definition

Provides mapping between fields of particular MOI to management interface over which it can be reported.

5.2.1.4.6.2 Attributes

The SupportedInterface contains attributes used to identify which interface field(s) are used to match the mapping.

Attribute name	S	isReadable	isWritable	isInvariant	isNotifiable
managementInterfaceId	M	T	F	T	F
eventToDictionaryMap	M	T	F	T	F

5.2.1.4.6.3 Attribute constraints

None

5.2.1.4.6.4 Notifications

This class does not support any notification.

5.2.1.4.6.5 State diagram

None.

5.2.1.4.7 EventToDictionaryMap <<choice>>

5.2.1.4.7.1 Definition

The EventToDictionaryMap is a choice stereotype represents one of a set of data types as shown per dictionary type in figures 5.2.1.3.1-1 and 5.2.1.3.1-2.

5.2.1.4.7.2 Attributes

As a choice stereotype the class has no intrinsic attributes. Instead, the attributes of the class are specified in the alternative class type definitions.

5.2.1.4.7.3 Attribute constraints

None

5.2.1.4.7.4 Notifications

This class does not support any notification.

5.2.1.4.7.5 State diagram

None.

5.2.1.4.8 PerformanceDictionaryMap <<dataType>>

5.2.1.4.8.1 Definition

The PerformanceDictionaryMap provide the attributes required to map a performance measure in a performance report to its corresponding PerformanceMeasureDefinition entry in an PerformanceMeasureDictionary. Performance measures for a given software product are reported over an interface.

A map shall be provided between the measurement report attribute and the performanceMeasureDefinitionId within the dictionary for a given interface. The PerformanceDictionaryMap provides the ability to identify the attribute name within a performance report which shall contain the value to use as the performanceMeasureDefinitionId in a PerformanceMeasureDictionary.

5.2.1.4.8.2 Attributes

The PerformanceDictionaryMap contains attributes used to identify which interface field(s) are used to match the performanceMeasureDefinitionId of a PerformanceMeasureDefinition_ in the PerformanceMeasureDictionary.

Attribute name	S	isReadable	isWritable	isInvariant	isNotifiable
eventToMeasureDefinitionMap	M	T	F	T	F

5.2.1.4.8.3 Attribute constraints

None

5.2.1.4.8.4 Notifications

This class does not support any notification.

5.2.1.4.8.5 State diagram

None.

5.2.1.4.9 Void

5.2.1.4.10 AlarmDictionary

5.2.1.4.10.1 Definition

There is more information about an alarm than is included in a notification about an alarm occurrence. The `AlarmDictionary` provides that data to the SMO such that the extended data may be used to assist in alarm event processing. These could include repair actions which guide a user of the SMO of what to do when the alarm occurs. If the alarm is based on a standard, this could also include the standard-defined descriptions/discussion about the alarm condition.

Along with Network Functions, O-Cloud resources or deployments may generate an alarm notification which are sent to subscribed consumers. The SMO shall be the primary consumer of these events which may need the ability to correlate them when there is a single root cause for alarms from multiple sources. The dictionary definition provides extensibility where such end-to-end type of signatures may be described.

Not all alarms definitions are standardized and therefore the alarm dictionary should be structured to allow a resource and/or application vendors to specify their unique alarms while also supporting alarms which are defined by a standard.

5.2.1.4.10.2 Attributes

The `AlarmDictionary` includes attributes inherited from `Top_ [2]` and the following attributes:

Attribute name	S	isReadable	isWritable	isInvariant	isNotifiable
<code>alarmDictionaryId</code>	M	T	F	T	F
<code>alarmDictionaryVersion</code>	M	T	F	T	F
<code>alarmDictionarySchemaVersion</code>	M	T	F	T	F
<code>supportedEntityTypes</code>	M	T	F	T	F
<code>vendor</code>	M	T	F	T	F
<code>supportedProducts</code>	M	T	F	T	F
<code>supportedInterfaces</code>	M	T	F	T	F
<code>alarmDefinitions</code>	M	T	F	T	F
<code>probableCauses</code>	O	T	F	T	F

5.2.1.4.10.3 Attribute constraints

None

5.2.1.4.10.4 Notifications

This class does not support any notification.

5.2.1.4.10.5 State diagram

None

5.2.1.4.11 AlarmDictionaryMap <<data Type>>

5.2.1.4.11.1 Definition

The AlarmDictionaryMap provides the attributes to map an alarm event record to its corresponding AlarmDefinition entry in an AlarmDictionary. Alarm event records for a given software product are reported over an interface. A map is required to identify the AlarmEvent attribute which correlates to the alarmDefinitionId within the dictionary for a given interface. The AlarmDictionaryMap identifies the attribute name within an Alarm event record, which shall contain the alarmDefinitionId value in an AlarmDictionary.

5.2.1.4.11.2 Attributes

The AlarmDictionaryMap defines following attribute:

Attribute name	S	isReadable	isWritable	isInvariant	isNotifiable
eventToAlarmDefinitionMap	M	T	F	T	F

5.2.1.4.11.3 Attribute constraints

None

5.2.1.4.11.4 Notifications

This class does not support any notification.

5.2.1.4.11.5 State diagram

None.

5.2.1.4.12 AlarmDefinition <<data Type>>

5.2.1.4.12.1 Definition

The AlarmDefinition includes additional static data about an alarm that is not transmitted in the Alarm event notification. The data is used by the consumer for alarm planning, automation, troubleshooting or to enrich the infrastructure alarm event notification when received.

5.2.1.4.12.2 Attributes

The AlarmDefinition inherits attributes from Top_ [2] and extends it with the following attributes:

Attribute name	S	isReadable	isWritable	isInvariant	isNotifiable
alarmDefinitionId	M	T	F	T	F
alarmName	M	T	F	T	F
alarmLastChange	M	T	F	T	F
alarmChangeType	M	T	F	T	F
alarmDescription	M	T	F	T	F
proposedRepairActions	CO	T	F	T	F
proposedRepairActionRefs	CO	T	F	T	F
clearingType	M	T	F	T	F
standardReference	M	T	F	T	F
managementInterfaceId	M	T	F	T	F
extensions	M	T	F	T	F

5.2.1.4.12.3 Attribute constraints

Attribute name	Definition
proposedRepairActions proposedRepairActionRefs	Each defined alarm requires at least a proposed Repair Action defined by proposedRepairActions or proposedRepairActionRefs.

5.2.1.4.12.4 Notifications

This class does not support any notification.

5.2.1.4.12.5 State diagram

None

5.2.1.4.13 ProbableCauseInfo <<dataType>>

5.2.1.4.13.1 Definition

The ProbableCauseInfo is intended to help operators determine the root cause of an event.

There are industry suggested lists of probable causes that could be used. When used the ProbableCauseInfo definition should allow for an indication of such a source.

The ProbableCause(s) used by the vendor for alarm events are listed independently in the alarm dictionary. However, each alarm event should have a Probable Cause that is specific to that alarm instance.

5.2.1.4.13.2 Attributes

The ProbableCauseInfo is a structured data type consisting of the following attributes:

Attribute name	S	isReadable	isWritable	isInvariant	isNotifiable
probableCause	M	T	F	T	F
probableCauseDescription	M	T	F	T	F
standardReference	M	T	F	T	F

5.2.1.4.13.3 Attribute constraints

None

5.2.1.4.13.4 Notifications

This class does not support any notification.

5.2.1.4.13.5 State diagram

None.

5.2.1.4.14 Product <<dataType>>

5.2.1.4.14.1 Definition

The `Product` identifies a vendor product and its version.

5.2.1.4.14.2 Attributes

The `Product` is a structured data type consisting of the following attributes:

Attribute name	S	isReadable	isWritable	isInvariant	isNotifiable
product	M	T	F	T	F
version	M	T	F	T	F

5.2.1.4.14.3 Attribute constraints

None

5.2.1.4.14.4 Notifications

This class does not support any notification.

5.2.1.4.14.5 State diagram

None.

5.2.1.5 Attribute definitions

Attribute Name	Documentation and Allowed Values	Properties
<code>3GPPPerformanceMeasurementDefinition.collectionMethod</code>	This attribute identifies the collection method for an 3GPP based measurement definition. allowedValues: as defined in 3GPP TS 32.404 [13] clause 3.3(b).	type: <code>3GPPCollectionMethod</code> multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False
<code>3GPPPerformanceMeasurementDefinition.description</code>	This attribute contains an explanation of the measurement operation. allowedValues: as defined in 3GPP TS 32.404 [13] clause 3.3(a).	type: string multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False
<code>clause</code>	This attribute identifies the clause in a specification that defines a measurement definition. allowedValues: N/A.	type: string multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False

Attribute Name	Documentation and Allowed Values	Properties
condition	This attribute identifies the condition in which the measurement will be updated. allowedValues: as defined in 3GPP TS 32.404 [13] clause 3.3(c).	type: string multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False
ETSIPerformanceMeasurementDefinition.collectionMethod	This attribute identifies the collection method for an ETIS based measurement definition. allowedValues: as defined in ETSI NFV IFA027 clause 5(b) [14]	type: ETSICollectionMethod multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False
ETSIPerformanceMeasurementDefinition.description	This attribute contains the description of the performance measurement. allowedValues: as defined in ETSI NFV IFA027 clause 5(a) [14]	type: string multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False
generation	This attribute identifies the cellular technology generation(s) that a performance measurement definition is applicable to. allowedValues: as defined in 3GPP TS 32.404 [13] clause 3.3(h).	type: 3GPPGeneration multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False
managementInterfaceId	This attribute identified the management interface that a performance measurement definition can be reported on. allowedValues: As defined in ManagementInterface	type: ENUM ManagementInterface multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False
measurementName	This attribute is a descriptive name of the measurement type. allowedValues: as defined in 3GPP TS 32.404 [13] clause 3.3.	type: string multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False
measurementResult	This attribute is a description of expected result value(s). allowedValues: as defined in 3GPP TS 32.404 [13] clause 3.3(d).	type: string multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False
measurementType	This attribute describes a short form of the measurement name specified in the header, which is used to identify the measurement type. allowedValues: as defined in 3GPP TS 32.404 [13] clause 3.3(e).	type: String multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False
measurementUnit	This attribute describes the unit of the measurement value. allowedValues: as defined in ETSI NFV IFA027 clause 5(d) [14]	type: multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False
measureObjectClass	This attribute describes the object class. allowedValues: as defined in 3GPP TS 32.404 [13] clause 3.3(f).	type: string multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False

Attribute Name	Documentation and Allowed Values	Properties
performanceDictionarySchemaVersion	This attribute defines the version of the schema version used to create the measurement definitions. allowedValues: N/A	type: string multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False
performanceDictionaryVersion	This attribute defines the version of the dictionary itself. allowedValues: N/A	type: multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False
PerformanceMeasureDefinition.supportedInterfaces	This attribute when present this defines the specific subset of interfaces over which the measurement is transmitted and may define an override of the interface key field mapping. If omitted, then the measure is transmitted over all interfaces defined in the dictionary level map. allowedValues: N/A	type: SupportedInterface multiplicity: 0..* isOrdered: True isUnique: N/A defaultValue: None isNullable: True
performanceMeasureDefinitionId	This attribute identifies a single performanceMeasurementDefinition within the dictionary. allowedValues: N/A	type: string multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False
PerformanceMeasureDictionary.supportedInterfaces	This attribute provides the list over which the performance measurements will be transmitted over. Not all performance measures are required to be over all supported interfaces. allowedValues: N/A	type: SupportedInterface multiplicity: 1..* isOrdered: True isUnique: N/A defaultValue: None isNullable: False
performanceMeasureId	This attribute identifies a single performanceMeasurementDefinition with a numeric value which may be used during serialization over a streaming interface to support data compaction. allowedValues: unsigned integer	type: Uint multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False
eventToDictionaryMap	This attribute is a placeholder for an alternative set of fields based on a choice. This field shall be replaced with those fields identified in the appropriate alternative of the choice.	type: EventToDictionaryMap multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False
eventToMeasureDefinitionMap	This attribute identifies the field attribute in a performance report which shall contain the value to be used as the measureDefinitionID in an PerformanceMeasureDictionary. AllowedValues: Any attribute name within a performance report.	type: String multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False
product	This attribute identifies the software product name provided by a vendor. allowedValues: N/A	type: String multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False

Attribute Name	Documentation and Allowed Values	Properties
purpose	This attribute, when supplied, describes who will be using the measurement. allowedValues: as defined in 3GPP TS 32.404 [13] clause 3.3(i).	type: String multiplicity: 0..1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: True
standardReference	This attribute, when provided, identifies a Standard where a performance measure is defined. allowedValues: N/A	type: StandardReference multiplicity: 0..1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: True
standardsDefinitionOrganization	This attribute identifies the Standards definition organization such as but not limited to ETSI or 3GPP. allowedValues: N/A	type: String multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False
standardSpecification	This attribute defines the Standard's title which in contains the measurement definition. allowedValues: N/A.	type: String multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False
supportedMeasures	This attribute provides the list of measurement definitions contained within a performance dictionary. allowedValues: N/A	type: 3GPPPerformanceMeasureDefinition or ETSIPerformanceMeasureDefinition multiplicity: 1..* isOrdered: True isUnique: N/A defaultValue: None isNullable: False
supportedProducts	This attribute provides the list of products and their versions that a vendor supports in a dictionary. allowedValues: N/A	type: Product multiplicity: 1..* isOrdered: True isUnique: True defaultValue: None isNullable: False
switchingTechnology	This attribute defines the switching domain(s) this measurement is applicable to. allowedValues: as defined in 3GPP TS 32.404 [13] clause 3.3(g).	type: String multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False
trigger	This attribute describes the trigger which causes the counter to be updated. allowedValues: as defined in ETSI NFV IFA027 clause 5(c) [14]	type: String multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False
vendor	This attribute identifies vendor name of a software product that the specified dictionary is provided for. allowedValues: as defined in clause 4.4.1 of 3GPP TS 28.622 [2]	type: String multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False

Attribute Name	Documentation and Allowed Values	Properties
version	This attribute identifies specific version of a vendor software product that the specified dictionary is provided for. allowedValues: N/A	type: String multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False
versionOrRelease	This attribute identifies the specification version, release, or date of publication, that identifies a specific instance of the specification. allowedValues: N/A	type: String multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False
alarmChangeType	This indicates the type of last change that occurred during the alarm last change. allowedValues: As defined by AlarmChangeType	type: ENUM AlarmChangeType multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False
alarmDefinitions	This attribute contains the list of alarm definitions which are or have been supported by the entity type and are ordered ascending by alarmDefinitionId. allowedValues: N/A	type: AlarmDefinition multiplicity: 1..* isOrdered: True isUnique: N/A defaultValue: None isNullable: False
alarmDefinitionId	This identifies a specific alarmDefinition instance in the AlarmDictionary. This is the Primary Key into the alarmDefinitions. NOTE: Multiple values, separated by “:” can be concatenated, according to the order defined by the list in eventToAlarmDefinitionMap allowedValues: N/A	type: String multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False
AlarmDefinition.managementInterfaceId	This attribute, defines the specific subset of interfaces over which the alarm is transmitted. allowedValues: As defined by ManagementInterface	type: ENUM ManagementInterface multiplicity: 1..* isOrdered: False isUnique: True defaultValue: None isNullable: False
alarmDescription	This provides a longer descriptive meaning of the alarm condition and a description of the consequences of the alarm condition. This is intended to be read by an operator to give an idea of what happened and a sense of the effects, consequences, and other impacted areas of the system. allowedValues: N/A	type: String multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False
AlarmDictionary.supportedInterfaces	This attribute provides the i/f list over which the alarms will be transmitted. allowedValues: N/A	type: SupportedInterface multiplicity: 1..* isOrdered: False isUnique: True defaultValue: None isNullable: False
alarmDictionarySchemaVersion	This attribute defines the schema version used to create the dictionary. allowedValues: N/A	type: String multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False

Attribute Name	Documentation and Allowed Values	Properties
alarmDictionaryVersion	This attribute defines the version of the dictionary itself. allowedValues: N/A	type: String multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False
alarmLastChange	This defines the date and time of the last change to an alarm definition. allowedValues: N/A	type: DateTime multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False
alarmName	This describes the human readable name of the alarm. allowedValues: N/A	type: String multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False
clearingType	This Indicates the type of clearing that is supported by a defined alarm. allowedValues: As defined by AlarmClearingType.	type: ENUM AlarmClearingType multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False
supportedEntityTypes	This attribute identifies the list of the type of entities that can generate alarms. Allowed values are not limited in order to allow for extensibility of the O-Cloud Resource Types.	type: string multiplicity: 0..* isOrdered: False isUnique: True defaultValue: None isNullable: True
eventToAlarmDefinitionMap	This attribute identifies the attribute(s) in an alarm event which contain the value(s) to be used as the alarmDefinitionID in an AlarmDictionary (e.g. for O2 i/f, the O2 alarmDefinitionId as defined in ORAN WG6 [i.40], for M-plane i/f, the M-plane faultID as defined in ORAN WG4 [i.1], for O1 i/f, the O1 alarmType, probableCause and optionally specificProblem as defined in O-RAN WG10 [8]). NOTE: The attribute order in the list defines the value order in the alarmDefinitionID attribute	type: String multiplicity: 1..* isOrdered: False isUnique: True defaultValue: None isNullable: False
extensions	This attribute provides a list of key value pairs denoting extended attributes of the Information Object Class (e.g. vendor specific values per standard attribute(s) of an alarm event record) allowedValues: N/A	type: NameValuePair multiplicity: 0..* isOrdered: False isUnique: True defaultValue: None isNullable: True
probableCause	This identifies a specific probableCause instance in the AlarmDictionary. This is the Primary Key into the ProbableCauseInfo dataType. allowedValues: for WG10 O1 Alarms as defined in 3GPP TS 28.111[16], clause 7.4 and for WG6 O-Cloud Alarms as defined in O-RAN.WG6.TS.O2IMS-INTERFACE [12], clause 4. NOTE: allowedValues can be extended for vendor-specific alarms.	type: String or Integer multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: False

Attribute Name	Documentation and Allowed Values	Properties
probableCauseDescription	This provides any additional information beyond the probableCause to describe the probable cause	type: String multiplicity: 0..1 isOrdered: N/A isUnique: N/A defaultValue: None isNullable: True
probableCauses	This attribute contains the list of probable causes which may be referenced in an alarm definition. allowedValues: N/A	type: ProbableCauseInfo multiplicity: 0..* isOrdered: False isUnique: True defaultValue: None isNullable: True
proposedRepairActions	This identifies one or more proposed actions for the defined alarm. allowedValues: N/A	type: String multiplicity: 0..* isOrdered: False isUnique: True defaultValue: None isNullable: True
alarmDictionaryId	This attribute serves as the identifier of an AlarmDictionary. allowedValues:N/A	type: String multiplicity: 1 isOrdered: False isUnique: True defaultValue: None isNullable: False
performanceDictionaryId	This attribute serves as the identifier of a PerformanceMeasureDictionary. allowedValues:N/A	type: String multiplicity: 1 isOrdered: False isUnique: True defaultValue: None isNullable: False
proposedRepairActionRefs	This identifies one or more proposed actions by reference for the defined alarm. It provides the possibility to refer to an external document(s), where it is possible to provide more long, complex and structured definition of repair action(s) allowedValues: N/A	type: String multiplicity: 0..* isOrdered: False isUnique: True defaultValue: None isNullable: True

6 O-RAN Data Models

6.1 Formal relationship (traceability) between O-RAN Data Models and the O-RAN Information Model

O-RAN Information Model development should precede and serve as the basis for the Data Model development.

As the modeling practices and processes within O-RAN mature, the Information Model and Data Models should co-evolve to develop the APIs required by specific use cases.

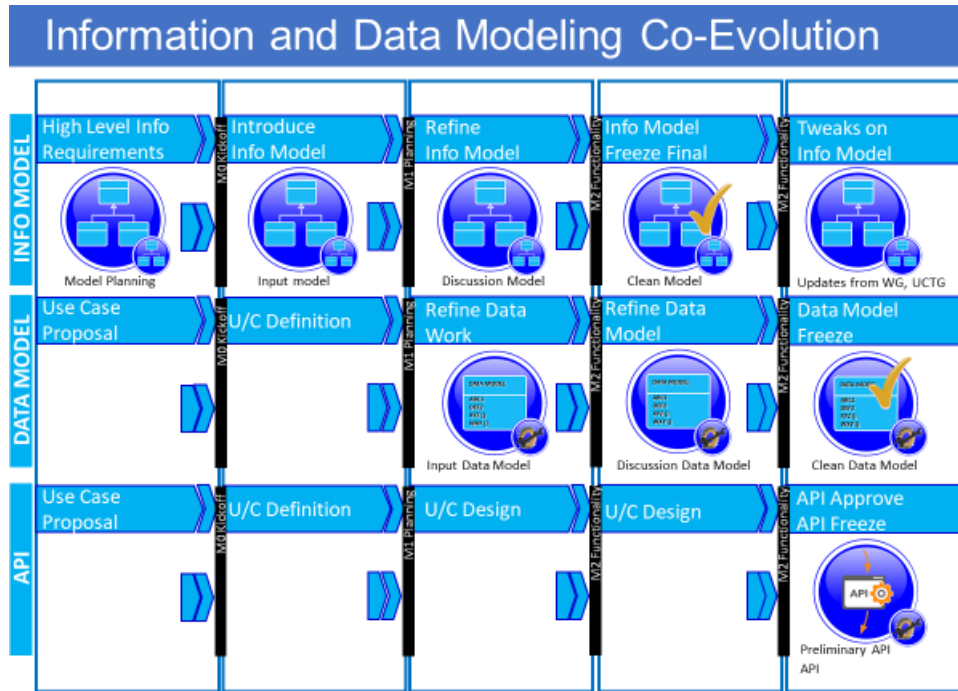


Figure 6.1-1 Information and Data Modeling Co-Evolution [i.3]

6.2 YANG Conventions

6.2.1 General

This clause describes the recommended conventions to be used in the O-RAN Alliance when writing YANG models.

In particular, because the creation and maintenance of YANG models is expected to be distributed across different working groups, this guide is intended to ensure that the way the models are organized and presented will be consistent across the entirety of the O-RAN Alliance.

6.2.2 Naming

MODULE FILE NAMING: All YANG modules shall have filenames that are unique within O-RAN. All YANG module filenames shall follow the form defined in clause 5.2 of IETF RFC 7950 [5], except for the revision date that shall not be present.

MODULE NAMING: All YANG module names shall have the form: “<org>-<identifier>-<name>” where:

- The <org> element is mandatory and shall have the value of “o-ran”.
- The <identifier> element is optional for YANG modules that have been already published. For new YANG modules the <identifier> element is mandatory. All YANG modules shall have <identifier> elements that are unique within O-RAN (e.g., o1).
- The <name> element is used to ensure the name of the YANG module is unique within the <org> or <org>-<identifier> module name conventions. Additionally in cases where a YANG module is dedicated to one O-RAN entity, the <name> element should contain name of this O-RAN entity, i.e. <name> = “odu-unique-name”, <name>

= “ocu-`<unique-name>`”. For O-RAN common YANG modules name should follow the form of `<name>`=“common-`<unique-name>`”.

MODULE NAMESPACE: All YANG modules shall have a namespace defined of the form namespace `"urn:<org>:<modulename>"`.

- The `<org>` element is mandatory and shall have the value of “o-ran”.
- The `<modulename>` element is mandatory and shall follow the module naming rules defined in this document.

PREFIX NAMING: Each YANG module shall have a prefix statement with a prefix that other dependent modules will use (also used in path references within the same module). YANG module prefixes shall be a maximum of 13 characters. All YANG modules shall have YANG module prefixes that are unique within O-RAN. All YANG module prefixes shall have the form “`<org>-[<abbreviatedidentifier>-]<abbreviatedname>`”.

- The `<org>` element is mandatory and shall have the value of “or” for new YANG modules.
- The `<org>` element is mandatory and may have the value of “o-ran” or “or” for YANG modules that already have been published.
- The `<abbreviatedidentifier>` element is optional for YANG modules that have been already published. For new YANG modules the `<abbreviated>` element is mandatory. All YANG modules shall have `<abbreviatedidentifier>` elementst that are unique within O-RAN (e.g., o1).
- The `<abbreviatedname>` element is used to ensure the YANG module prefix is unique within the `<org>` or `<org>-<abbreviatedidentifier>` YANG module prefix conventions.

6.2.3 Revision Statement

YANG modules revision shall follow the 3GPP TS 32.160 [7] clause 6.2.1.13 where a YANG module shall contain a revision statement following RFC 7950 [5], section 7.1.9., but the revision statements contents are defined as follows. The revision statement in the YANG modules shall include a description substatement used to track the versioning of the YANG model. A version number shall be in a format X.Y.Z according to SemVer [i.39]. In addition, the revision statement shall also include a reference sub statement used to referred document (e.g. stage 2 Information Model).

For example:

```
revision 2024-07-10 {  
  description "1.2.0";  
  reference "O1NRM";  
}
```

6.2.4 Indents

O-RAN Alliance YANG models shall use two-space tab indents.

6.2.5 YANG Language Usage

YANG VERSION: All models shall use YANG data modeling language version 1.1 (RFC 7950 [5])

KEY-LESS LISTS: Although permitted in YANG, the use of a list without a defined key should be avoided.

VALIDATION: All YANG modules should be validated / compiled with pyang tool using the following flag: `pyang --lint <module>`. The pyang errors shall be removed and pyang warning should be considered for removal. Note, successful

compilation with pyang does not guarantee a working model, as xPATH expressions aren't evaluated and forbidden operational data dependencies in the configuration may not generate appropriate errors.

6.2.6 Cross Working Group Co-ordination

Models that are likely to be applicable to more than one O-RAN Alliance working group should provide clear delineation between separate working groups configuration and/or state. The use of feature and if-feature should be used to ensure that NETCONF servers are not required to implement the entire data model, e.g., when aspects of such relate to the individual working group defined use cases. The feature name should indicate which working group or which O-RAN entity the capabilities have been defined by.

6.2.7 Void

6.2.8 Augmentations of YANG Data Models

O-RAN shall not modify other SDO modules.

O-RAN augmentation to the other SDO YANG modules shall be made in separate O-RAN originated YANG Data Model file, as modification of original SDO YANG Data Model file is prohibited.

Vendor specific augmentation to O-RAN YANG modules shall be made in separate vendor originated YANG Data Model files.

Annex A (informative): Links to Data Models approved for use

From O-RAN

The public facing web page that includes the formally released versions of models that are approved for use:

www.o-ran.org/specifications

From 3GPP

3GPP is one of the sources for management plane specifications for O-RAN components. 3GPP publishes its “SA5 Data models” including its yang data models in a publicly available git repository:

<https://forge.3gpp.org/rep/sa5/MnS>

From IETF

IETF is a complementary source for management plane specifications for O-RAN components. IETF publishes its yang data models (inclusive of IANA yang data models) in a publicly available git repository:

<https://github.com/YangModels/yang/tree/master/standard/ietf>

Annex B (informative): Relations between Information Model and Data Models

B.1 Information and Data Models as a Modeling Continuum

Within O-RAN, the de facto methods and procedures with respect to the early stage of an O-RAN “modeling continuum” aim to evolve **one common and coherent Information Model** for providing O-RAN extensions to the existing 4G/5G IMs such as the 3GPP NRMs, from which O-RAN Data Models may be generated.

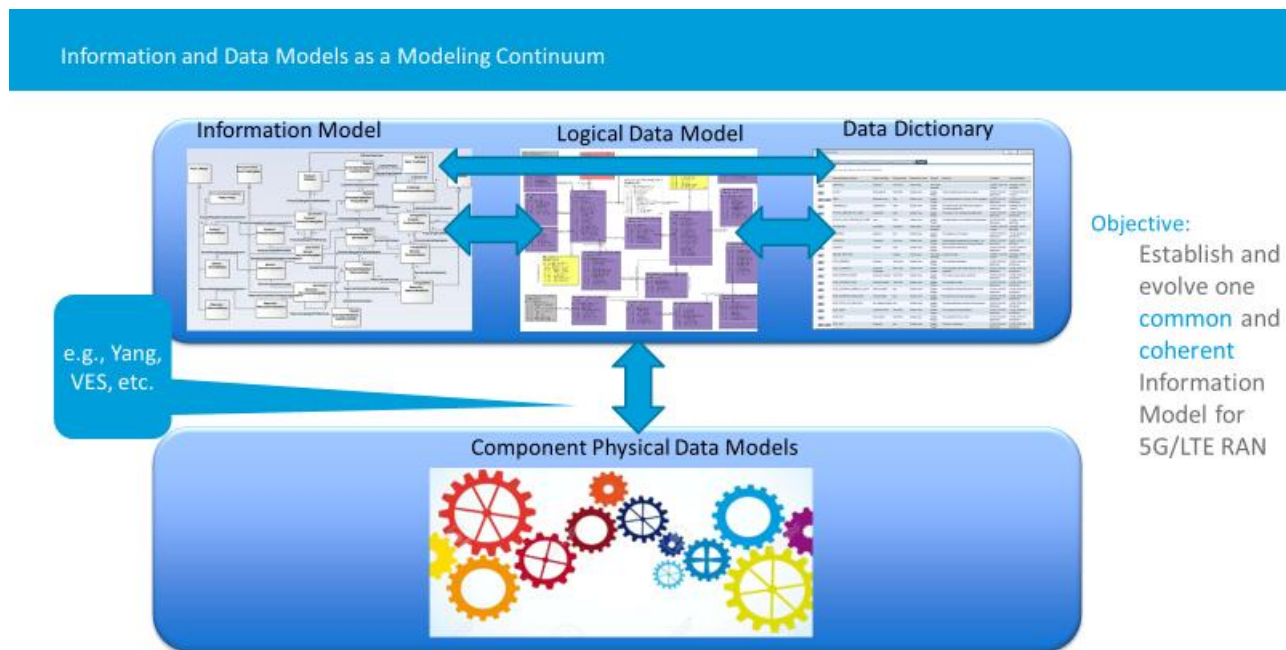


Figure B.1-1 Information and Data Models as a Modeling Continuum (conceptual)

B.2 Information and Data Modeling Co-Evolution

The “Modeling Continuum” depicted as Figure B.2-1 is purely conceptual and is intended to provide a framework for Information and Data Modeling Co-evolution.

Here is an example of the process to be adapted to, and adopted by, O-RAN:

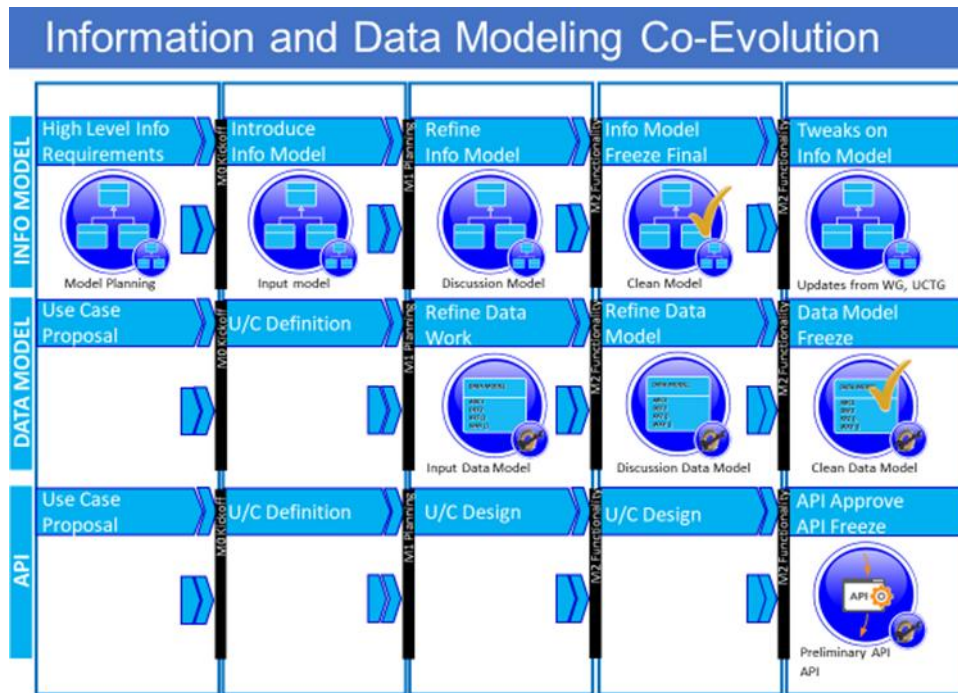


Figure B.2-1 Information and Data Modeling Co-Evolution [i.3]

B.3 Model and Use Case Development (process)

There is evolving process within O-RAN to guide and inform Information Model and Data Model(s) development based on prioritized use cases.

Here is example of the process to be adapted to, and adopted by, O-RAN.

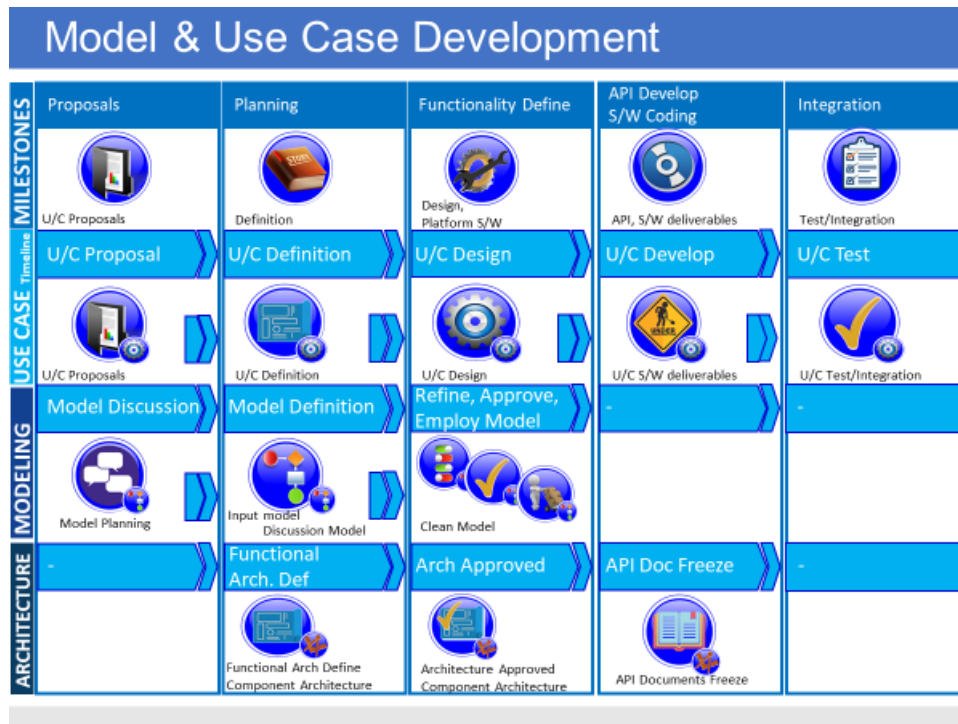


Figure B.3-1 Model and Use Case Development (process) [i.3]

Annex C (informative): Classes/components and interfaces that comprise the O-RAN Information Model and Data Models

Following are a list of classes/components as well as interfaces that may be part of the O-RAN Information model, along with the working group developing this entity, any SDO references if appropriate, followed by comments and the status. This is based on the premise that each WG is responsible for modeling the entity, and WG10 is responsible for stewarding the overarching model inclusive of the input from the other groups.

Entity	WG Developing Model	Doc Reference	Comments	Status
<i>Class / Component</i>				
NonRTRIC	WG2		Not present in model	
NearRTRIC	WG3		Presently shell only in model.	
O-CU-CP	WG5	O-RAN-WG5.O-CU-O1.0 [i.6]		
O-CU-UP	WG5	O-RAN-WG5.O-CU-O1.0 [i.6]		
O-DU	WG5	O-RAN.WG5.O-DU-O1.0 [i.6]		
O-RU	WG4		Shell only	
ManagedApplication	WG10		Abstract Class	
ManagedElement		3GPP TS 28.622 [2]	Class and attributes	
xApp	WG3, (WG6)		Not present in model	
rApp	WG2, (wg6)		Not present in model	
<i>Interfaces</i>				
A1-P	WG2	O-RAN.WG2.A1AP [i.4]		
A1-ML	WG2		Shell Only - to be pursued in a later release	
A1-EI	WG2		Shell Only - to be pursued in a later release	
O1	WG10	O-RAN.WG10.O1 Interface.0 [8]		
E1	WG5	3GPP TS 38.460 [i.10]	3GPP start, including operations list	
E2	WG3	O-RAN.WG3.E2GAP [i.5]	Interface with 9 operations	
F1-c	WG5	3GPP TS 38.470 [i.11]	3GPP start, including 24 operations	

F1-u	WG5	3GPP TS 38.470 [i.11]	3GPP start, including limited operation list	
OpenFrontHaul	WG4	O-RAN.WG4.TS.MP.0 [i.1]	Shell only	
X2	WG5	3GPP TS 36.423 [i.9]	3GPP based start, including a large number of operations	
O2	WG6		Not present in model	
R1	WG2		Not present in model	

Table C-1 Classes/components and interfaces that comprise the O-RAN Information Model and Data Model(s)

Annex D (informative): Usage of models

D.1 Usage of 3GPP Data Models

The O-RAN Alliance complements the work of other SDOs. 3GPP is the primary source for management plane specifications for O-RAN components.

Please refer to Annex A for the 3GPP-source Data Models that are approved for use.

The following table (snapshot) shows a subset of the 3GPP yang data models to an O-RAN model construct.

3GPP yang data model	3GPP spec for YANG model	o-ru (O1 in hybrid mode currently not supported)	o-du	o-cu-up	o-cu-cp	near-rt-ric
_3gpp-5g-common-yang-types	3GPP TS 28.541 [i.37]	imported by _3gpp-common-managed-element (3GPP TS 28.623 [i.8])	imported by _3gpp-common-managed-element (3GPP TS 28.623 [i.8])	imported by _3gpp-common-managed-element (3GPP TS 28.623 [i.8])	imported by _3gpp-common-managed-element (3GPP TS 28.623 [i.8])	imported by _3gpp-common-managed-element (3GPP TS 28.623 [i.8])
_3gpp-common-ep-rp	3GPP TS 28.623 [i.8]	[open] o-ran-m-int.yang & o-ran-ru-rtf.yang defines the interface of O-RU, maybe it is not needed to O-RU	imported by _3gpp-nr-nrm-ep (3GPP TS 28.541 [i.37]) abstract superclass for all 3GPP endpoints [open] as O-RU remote PORT, to configure eCPRI port of O-DU just EP_RP looks not enough	imported by _3gpp-nr-nrm-ep (3GPP TS 28.541 [i.37]) abstract superclass for all 3GPP endpoints	imported by _3gpp-nr-nrm-ep (3GPP TS 28.541 [i.37]) abstract superclass for all 3GPP endpoints	[open] need discussion , if RIC modeled as O-RU which need detailed configuration to interfaces, the common part probably from IETF
_3gpp-common-fm	3GPP TS 28.623 [i.8]	imported by _3gpp-common-managed-element (3GPP TS 28.623 [i.8]) needed for alarm list handling	imported by _3gpp-common-managed-element (3GPP TS 28.623 [i.8]) needed for alarm list handling	imported by _3gpp-common-managed-element (3GPP TS 28.623 [i.8]) needed for alarm list handling	imported by _3gpp-common-managed-element (3GPP TS 28.623 [i.8]) needed for alarm list handling	imported by _3gpp-common-managed-element (3GPP TS 28.623 [i.8]) needed for alarm list handling
_3gpp-common-managed-element	3GPP TS 28.623 [i.8]	root object class	root object class	root object class	root object class	root object class
_3gpp-common-managed-function	3GPP TS 28.623 [i.8]	needed to extend MF for O-RU functionality	imported by _3gpp-nr-nrm-gnbdufunction (3GPP TS 28.541 [i.37])	imported by _3gpp-nr-nrm-gnbdufunction (3GPP TS 28.541 [i.37])	imported by _3gpp-nr-nrm-gnbdufunction (3GPP TS 28.541 [i.37])	needed either for a standalone RIC or combined RIC ME

_3gpp-common-measurements	3GPP TS 28.623 [i.8]	imported by _3gpp-common-managed-element (3GPP TS 28.623 [i.8]) needed for PM job control and threshold monitoring	imported by _3gpp-common-managed-element (3GPP TS 28.623 [i.8]) needed for PM job control and threshold monitoring	imported by _3gpp-common-managed-element (3GPP TS 28.623 [i.8]) needed for PM job control and threshold monitoring	imported by _3gpp-common-managed-element (3GPP TS 28.623 [i.8]) needed for PM job control and threshold monitoring	imported by _3gpp-common-managed-element (3GPP TS 28.623 [i.8]) needed for PM job control and threshold monitoring
_3gpp-common-subnetwork	3GPP TS 28.623 [i.8]	imported by _3gpp-common-subscription-control (3GPP TS 28.623 [i.8]) not needed from O-RAN point of view but can't be removed with modification of the yang, Contribution to 3GPP required	imported by _3gpp-common-subscription-control (3GPP TS 28.623 [i.8]) not needed from O-RAN point of view but can't be removed with modification of the yang, Contribution to 3GPP required	imported by _3gpp-common-subscription-control (3GPP TS 28.623 [i.8]) not needed from O-RAN point of view but can't be removed with modification of the yang, Contribution to 3GPP required	imported by _3gpp-common-subscription-control (3GPP TS 28.623 [i.8]) not needed from O-RAN point of view but can't be removed with modification of the yang, Contribution to 3GPP required	imported by _3gpp-common-subscription-control (3GPP TS 28.623 [i.8]) not needed from O-RAN point of view but can't be removed with modification of the yang, Contribution to 3GPP required
_3gpp-common-subscription-control	3GPP TS 28.623 [i.8]	needed for VES subscription	needed for VES subscription	needed for VES subscription	needed for VES subscription	needed for VES subscription
_3gpp-common-top	3GPP TS 28.623 [i.8]	imported by _3gpp-common-managed-element (3GPP TS 28.623 [i.8]) abstract class supplying a naming attribute	imported by _3gpp-common-managed-element (3GPP TS 28.623 [i.8]) abstract class supplying a naming attribute	imported by _3gpp-common-managed-element (3GPP TS 28.623 [i.8]) abstract class supplying a naming attribute	imported by _3gpp-common-managed-element (3GPP TS 28.623 [i.8]) abstract class supplying a naming attribute	imported by _3gpp-common-managed-element (3GPP TS 28.623 [i.8]) abstract class supplying a naming attribute
_3gpp-common-yang-extensions	3GPP TS 28.623 [i.8]	expected for inVariant	expected for inVariant	expected for inVariant	expected for inVariant	expected for inVariant
_3gpp-common-yang-types	3GPP TS 28.623 [i.8]	imported by many other 3GPP yang modules essential 3GPP typedefs, in particular DistinguishedName, and other useful typedefs like OperationalState, AdministrativeState, AvailabilityStatus	imported by many other 3GPP yang modules essential 3GPP typedefs, in particular DistinguishedName, and other useful typedefs like OperationalState, AdministrativeState, AvailabilityStatus	imported by many other 3GPP yang modules essential 3GPP typedefs, in particular DistinguishedName, and other useful typedefs like OperationalState, AdministrativeState, AvailabilityStatus	imported by many other 3GPP yang modules essential 3GPP typedefs, in particular DistinguishedName, and other useful typedefs like OperationalState, AdministrativeState, AvailabilityStatus	imported by many other 3GPP yang modules essential 3GPP typedefs, in particular DistinguishedName, and other useful typedefs like OperationalState, AdministrativeState, AvailabilityStatus

Table D.1-2 Mapping of 3GPP yang data models to O-RAN element functions

D.2 Usage of non-3GPP data models

There are domains of data-models being considered by O-RAN WGs (Working Groups) that are not covered by 3GPP but are covered by other SDOs, e.g., IETF, MEF, IEEE, ONF, BBF, and occasionally imported by 3GPP.

The following table shows data models that are of interest and being considered within O-RAN and/or 3GPP.

Order No	yang data model	Specification	o-ru (M-Plane)	o-du	o-cu-up	o-cu-cp	near-rt-ric	Comments
001	ietf-yang-types	RFC 6991 [i.18]	import by several models	import by _3gpp-common-yang-types.yang (3GPP TS 28.623 [i.8])	import by _3gpp-common-yang-types.yang (3GPP TS 28.623 [i.8])	import by _3gpp-common-yang-types.yang (3GPP TS 28.623 [i.8])	import by _3gpp-common-yang-types.yang (3GPP TS 28.623 [i.8])	
002	ietf-inet-types	RFC 6991 [i.18]	import by several models	import by _3gpp-common-yang-types.yang (3GPP TS 28.623 [i.8])	import by _3gpp-common-yang-types.yang (3GPP TS 28.623 [i.8])	import by _3gpp-common-yang-types.yang (3GPP TS 28.623 [i.8])	import by _3gpp-common-yang-types.yang (3GPP TS 28.623 [i.8])	
011	ietf-netconf.yang	RFC6241 [i.2]	Used	used	used	used	used	
012	ietf-netconf-acm.yang	RFC8341 [i.24]	baseline for access control					Network Configuration Access Control Model
013	ietf-netconf-monitoring	RFC6022 [i.14]	used					NETCONF Monitoring Module

014	ietf-netconf-nmda	RFC8526 [i.31]						NETCONF operations to support the Network Management Datastore Architecture
015	ietf-netconf-notifications	RFC6470 [i.16]	used	not O1 but OpenFronthaul mPlane, O-RAN.WG4.TS.MP.0 [i.1]				This module defines a YANG data model for use with the NETCONF protocol that allows the NETCONF client to receive common NETCONF base event notifications
016	ietf-netconf-partial-lock	RFC5717 [i.13]						
017	ietf-netconf-time	RFC7758 [i.21]						time-triggered configuration and management operations
018	ietf-netconf-with-defaults	RFC6243 [i.15]						NETCONF client to control how default values are handled by the server in particular NETCONF operations

020	ietf-system	RFC7317 [i.20]	overlaps with o-ran-operations but may be required to support 802.1X					configuration and identification of some common system properties within a device containing a NETCONF server - time-zone management - radius - local users - NTP
030	ietf-hardware	RFC8348 [i.28]	used	not O1 but OpenFronthaul mPlane, O-RAN.WG4.TS.MP.0 [i.1]				
031	iana-hardware	RFC8348 [i.28]	import by ietf-hardware	import by ietf-hardware				IANA-defined identities for hardware class.
032	ietf-hardware-state	RFC8348 [i.28]						hardware monitoring
033	ietf-interfaces	RFC8343 [i.26]	foundation for fronthaul					
034	iana-if-type	RFC7224 [i.19]	used	not O1 but OpenFronthaul mPlane, O-RAN.WG4.TS.MP.0 [i.1]				YANG identities for IANA-registered interface types

040	ietf-alarms	RFC8632 [i.34]						This module defines an interface for managing alarms.
	ietf-ip	RFC8344 [i.27]	foundation for fronthaul M-Plane, O-RAN.WG4 .TS.MP.0 [i.1]					managing IP implementations
	ietf-ptp	RFC8575 [i.33]	considered - but decided o define ownno-ran-sync.yang					configuration of IEEE Std 1588-2008 clocks
	ietf-yang-library	RFC8525 [i.30]	foundation for YANG 1.1 (RFC 7950 [5])					
	ietf-yang-metadata	RFC7952 [i.22]						This YANG module defines an 'extension' statement that allows for defining metadata annotations
	ietf-yang-patch	RFC8072 [i.23]						This module contains conceptual YANG specifications for the YANG Patch and YANG Patch Status data structures.
	ietf-yang-push	RFC8641 [i.35]						

	ietf-yang-schema-mount	RFC8528 [i.32]						This module defines a YANG extension statement that can be used to incorporate data models defined in other YANG modules in a module.
	ietf-yang-smiv2	RFC6643 [i.17]						This module defines YANG extensions that are used to translate SMiv2 concepts into YANG.
	iana-crypt-hash	RFC7317 [i.20]	used in CTI					
	ietf-crypto-types	draft-ietf-netconf-crypto-types [i.36]	used by fronthaul file management					
	ieee802-dot1x	IEEE 802.1X-2020 [i.12]	being considered by STG					

Table D.2-3 Non 3GPP yang models

Annex (informative): Change History

Date	Revision	Description
2020.11.25	01.00	First draft of O-RAN Information Model and Data Models
2021.11.19	02.00	Extension of YANG style guide with respect to O1 (3GPP) aligned models creation. Alignment of the document to respect the fact that it is owned by WG10. Addition of a link to WG10 repository Further editorial fixes.
2022.03.30	03.00	Editorial corrections References updates Papyrus installation description update Abbreviations update
2022.11.09	04.00	Guidance on how to create IMs provided Introduction of Managed Application IOC IM Removal of Anex ZZZ Updated copyright statement Updated references
2023.03.27	05.00	Performance Dictionary IM for O2 IMS provided Fault Dictionary IM for O2 IMS provided Alignment of descriptive text with OAM Architecture Removal of Papyrus installation clause Cleanup of text
2023.07.17	06.00	References cleanup Clause renumbering Figures renumbering 3GPP references update
2023.10.30	07.00	ETSI PAS Alignment cleanup References Update
2024.07.26	09.00	Terms clause update - Removal of "Logical Data Model" - Removal of "Component Physical Data Models" - Addition of "Namespace" Model definitions update in clause 3.1 Namespace definition update Removing of O1 related NRMs from model Namespaces introduction Editorial fixes

2024.12.06	10.00	Attributes performanceDictionaryId and alarmDictionaryId added Section 6.2.2 Naming: - Update naming conventions to recommendations. - Add prefix naming convention that includes the short interface name. Change 3GPP references to Rel 18. Editorial fixes
2025.03.10	11.00	Dictionary Support for multiple vendor products Improve mapping rule from Stage 2 to Stage 3
2025.07.21	12.00	Implemented CRs: - ERI-2025.03.25-O-RAN-CR-0162-ProposedRepairActionRefs-v03 - ERI-2025.04.01-O-RAN-CR-0163-AlarmDef-v04 Editorial fixes Alignment with new TS template
2025.11.24	13.00	Implemented CR: - DCM-2025.09.01-WG10-CR-0030-Correction for probableCause attribute description-v02 - NOK-2024.09.23.WG10-CR-0096 Dictionary attributes update-v01 Figures numbers fixes Further editorial fixes